Trabajo de Fin de Máster
Máster Universitario en Lógica, Computación e IA

A Study on Foundational Visual-Language Models for Histopathology Image Analysis

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Agradecimientos

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Leandro Candau Sánchez de Ybargüen
Sevilla, 2026

Resumen

- Contexto: clasificación de imágenes histopatológicas con datos etiquetados limitados y etiquetas repetidas.
- Enfoque: aprendizaje contrastivo imagen-texto al estilo CLIP con codificadores fundacionales congelados.
- Datos: Kaggle LC25000, 5 clases de tejido de pulmón y colon.
- Diseño experimental: línea base controlada + matriz de ablaciones (una variable a la vez).
- Resultados (parciales): línea base 63.6 % accuracy / 62.7 % macro-F1; PLIP cero-disparo +7.3 macro-F1; ajuste fino parcial (últimas 30 capas) +27.1 macro-F1 (89.8 % / 89.8 %).

Abstract

- Context: histopathology image classification under limited labelled data and repeated labels.
- Approach: CLIP-style image-text contrastive learning with frozen foundation-model encoders.
- Data: Kaggle LC25000, five lung and colon tissue classes.
- Experimental design: a controlled baseline + a single-variable ablation matrix.
- Results (partial): baseline 63.6 % accuracy / 62.7 % macro-F1; PLIP zero-shot +7.3 macro-F1; partial fine-tuning (last 30 layers) +27.1 macro-F1 (89.8 % / 89.8 %).

Contents

1 Introduction

Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision.

RADFORD ET AL., 2021

1.1 Motivation

- Clinical relevance of histopathology classification.
- Limited labelled data - recurring limitation in this domain.
- Foundation models - able to do different tasks with little task-specific supervision.
- Project question: "How does CLIP transfer to histopathology with different backbones? Where are its limits?"
- Framing: CLIP as the object of study, not the classifier. Findings should generalise to other scientific domains where short or missing labels are the norm.

1.2 Project Scope Evolution

- Initial direction -> semantic segmentation. Abandoned because annotated segmentation data was too scarce within the time budget.
- Pivot -> image-level classification, same domain, reframed question.
- Testbed: LC25000. Reference point: PLIP.

1.3 Research Questions

- RQ1: Frozen general-purpose CLIP transfer to LC25000 — quality and failure modes.
- RQ2: Pathology pretraining contribution (PLIP vs the general-purpose baseline).
- RQ3: Feature-adaptation contribution (partial unfreezing of the image encoder).
- RQ4: CLIP-paradigm contribution vs a plain image classifier on the same frozen features.
- RQ5: Stain normalization $\rightarrow$ shortcut-learning probe (Macenko / Reinhard / Vahadane head-to-head).
- RQ6: Biomedical text encoder (BioBERT / PubMedBERT) under short prompts.
- RQ7: Out-of-distribution transfer (external colon dataset).

1.4 Contributions

The central contribution of this project is a controlled measurement of how much each major design choice in a CLIP-style histopathology pipeline actually matters, with every claim framed as a delta over the same controlled baseline — frozen ResNet50 + frozen BERT, single fixed prompt, stratified 80/10/10 split on LC25000 — and iterated through one experiment in Chapter 4.

Measured claims:

- **Pathology-domain pretraining.** PLIP zero-shot vs the general-purpose baseline: $+7.3$ macro-F1 (69.8 % / 70.0 % vs 63.6 % / 62.7 %), with no LC25000 training. Gain concentrates on benign tissues; the two cancer subtypes (lung_aca, colon_aca) tie.
- **Feature-level adaptation.** Last 30 layers of ResNet50 unfrozen, LR dropped $10\times$ to 10^{-4}, everything else held constant: $+27.1$ macro-F1 (89.8 % / 89.8 %). Broad-based; every class $F_1 \geq 0.85$; lung_aca moves $0.48 \rightarrow 0.85$.
- **CLIP architecture vs plain CE classifier on the same features.** A 2×2 comparison at each adaptation budget. Frozen regime: tied within noise ($+1.0$ macro-F1 to plain CE, 0.6369 vs 0.6273). Unfreeze30 regime: CLIP wins $+2.6$ macro-F1 (0.8981 vs 0.8723), broad-based across classes. The CLIP architecture neither costs nor benefits accuracy at the frozen-features ceiling but gains a small consistent advantage when features adapt, plausibly via implicit regularisation from the InfoNCE loss.

Methodology:

- Single-variable-change protocol: each experiment differs from the baseline in exactly one design choice, so the resulting delta is attributable.
- Artifact bundle (fixed-seed split, weights, metric files, plots) versioned alongside the report in the project repository.
- Explicit written discussion of dataset and methodological limits constraining how far each delta can be generalised.

Pending ablations, scoped and implemented, to be folded in as measured claims once the corresponding run lands:

- Stain normalization head-to-head: Macenko, Reinhard, Vahadane vs raw inputs.
- Biomedical text encoder swap: BioBERT and PubMedBERT vs frozen BERT base.
- Out-of-distribution transfer of the baseline checkpoint to an external colon dataset.

1.5 Document Structure

- Chapter map to be added in the prose pass.

2 State of the Art

AI is undergoing a paradigm shift with the rise of models [...] that are trained on broad data at scale and are adaptable to a wide range of downstream tasks.

Bommasani et al., 2021

- Foundation models framing (Bommasani 2021).
- CLIP and multimodal contrastive learning (Radford 2021) — the paradigm under study.
- Self-supervised visual pretraining for medical imaging — DINO, DINOv2.
- Pathology shift: per-tile supervised CNNs $\rightarrow$ foundation models (2016–2024).
- Pathology-specific foundation models:
  - TransPath (2022) — transformer + SSL.
  - Phikon (2023) — masked image modelling.
  - PLIP (2023) — CLIP on medical Twitter.
  - CONCH (2024) — larger pathology VLM.
  - BiomedCLIP, MUSK — recent entrants.
- Open challenges: benchmarks, dataset bias, cross-institution generalization, compute scale.
- Where this project fits: small-scale empirical study of general-purpose foundation backbones on pathology classification, with single-variable ablations.

3 Methodology

3.1 Overview

- One-page chapter map + project question recap.
- Reading order: background → baseline → ablations.

3.2 Histopathology Digital Imaging

- H&E staining.
- Whole-Slide Imaging + patch extraction.
- Stain variability and normalization (Macenko, Reinhard, Vahadane).
- Why the domain is hard for general-purpose VLMs: domain gap from natural images, no per-image captions in clinical archives, dataset artifacts that can drive performance.

3.3 Contrastive Vision-Language Learning (CLIP)

3.3.1 Why CLIP?

- CLIP as object of study, not the classifier.
- Capabilities CLIP provides over a softmax head:
  - Open-vocabulary classification (new classes via prompts proven by adding new samples).
  - Semantic similarity output (cosine vs natural language).
  - Prompt engineering as a sensitivity lever.

3.3.2 Architecture and training signal

- Dual-encoder + shared embedding space.
- InfoNCE loss, learnable temperature.
- In-batch false-negative regime at small batch + few classes.
- Prompt-based zero-shot classification.
- CLIP-paper sample-level retrieval → doesn't apply (class-shared prompts).

3.4 Image Encoders

- ResNet (CNN baseline).
- Vision Transformer (ViT).
- DINO / DINOv2 (self-supervised).

Two independent encoders project image and text into a shared d-dimensional embedding space. The cosine-similarity matrix across a batch is the input to the symmetric InfoNCE loss.

Figure 3.1 CLIP dual-encoder architecture used in this project..

3.5 Text Encoders

- Transformer + BERT — text encoder used by baseline.
- BioBERT / PubMedBERT — biomedical text encoder swap ablation.

3.6 Reference baselines: PLIP and a plain image classifier

- PLIP zero-shot (Huang 2023) — pathology-pretrained CLIP, frozen, no LC25000 training.
- Plain image classifier — Dense(5)+softmax on frozen ResNet50 trained with cross-entropy; no text encoder, no CLIP architecture.
- Together: PLIP isolates the pretraining-domain contribution; the plain classifier isolates the CLIP-architecture contribution.

3.7 Dataset: Kaggle LC25000

- Borkowski 2019.
- 5 tissue classes (lung, colon).
- 25 000 patches from 1 250 originals via rotation/flip → no further geometric augmentation.
- Known limits: single institution, possible stain artifacts, no patient-level grouping.

Table 3.1 LC25000 class counts and stratified split sizes..

Class	**Total**	**Train**	**Val**	**Test**
benign lung tissue	5 000	4 000	500	500
lung adenocarcinoma	5 000	4 000	500	500
lung squamous cell carcinoma	5 000	4 000	500	500
benign colon tissue	5 000	4 000	500	500
colon adenocarcinoma	5 000	4 000	500	500
Total	25 000	20 000	2 500	2 500

Five-class sample: "benign lung", "lung adenocarcinoma", "lung squamous cell carcinoma", "benign colon", "colon adenocarcinoma".

Figure 3.2 Example image per class from LC25000..

3.8 Preprocessing and Input Pipeline

- Image: resize 224×224, ResNet-style preprocessing, no further augmentation.
- Text: BERT WordPiece, fixed sequence length.
- Split: stratified 80/10/10, seed 42, persisted.

3.9 The Baseline

- Frozen ResNet50 + frozen BERT.
- Projection heads (Dense + LayerNorm) $\rightarrow$ 256-d shared space.
- Learnable temperature, CLIP-style init.
- Symmetric InfoNCE, AdamW, grad clipping.
- Single fixed prompt at train + eval.
- Float32 throughout.

Table 3.2 Baseline hyperparameters. Ablations change exactly one row..

Hyperparameter	**Value**
Image encoder	ResNet50 (ImageNet preset), frozen
Text encoder	BERT base uncased, frozen
Projection head	Dense(256, no bias) $\rightarrow$ LayerNorm
Embedding dimension	256
Image size	224×224
Text sequence length	24 WordPiece tokens
Batch size	32
Initial temperature	$\tau = 0.07$
Loss	symmetric InfoNCE
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Gradient clipnorm	1.0
Max epochs	30
EarlyStopping patience	5
ReduceLROnPlateau	factor 0.5, patience 2, min 10^{-6}
Numerical precision	float32
Random seed	42

3.10 Evaluation Protocol

- Classification: argmax cosine(image, class-prompt).
- Global: accuracy, balanced accuracy, macro-F1, weighted-F1.
- Per-class: precision, recall, F1.
- Confusion matrices: raw + row-normalized.
- Embeddings: shared image+prompt UMAP, centroid-to-prompt heatmap.
- Class-level retrieval, with explicit positive definition.

3.11 Experimental Design and Ablation Protocol

- Single-variable-change principle.
- Matrix grouped by axis of variation; full matrix in §4.1.2.
- Scope boundaries: no patient-level grouping, no human-expert evaluation, no architecture-controlled comparison vs CONCH / BiomedCLIP / MUSK. Deferred to future work.

3.12 Computational Setup and Reproducibility

- Hardware: Google Colab T4, local fallback.
- Stack: TensorFlow + KerasHub; PyTorch + transformers for PLIP only.
- Pinned library versions.
- Seeds + PYTHONHASHSEED + `tf.config.experimental.enable_op_determinism()`.

4 Results and Discussion

4.1 Experimental Overview

4.1.1 Research Questions

- RQ1: Frozen general-purpose CLIP transfer — quality + failure modes.
- RQ2: Pathology pretraining contribution (PLIP vs baseline).
- RQ3: Feature adaptation contribution (partial unfreezing).
- RQ4: CLIP paradigm value vs a plain image classifier on the same frozen features.
- RQ5: Stain normalization → stain shortcut + Macenko / Reinhard / Vahadane head-to-head.
- RQ6: Biomedical text encoder (BioBERT / PubMedBERT) under short prompts.
- RQ7: OOD transfer to external colon dataset.

4.1.2 Experiment Matrix

- Grouped by axis of variation. Baseline is the reference; others change exactly one variable.
- All rows on same 80/10/10 split unless testing external generalization.

4.2 Baseline: frozen ResNet50 + frozen BERT

4.2.1 Configuration

- Frozen ResNet50 (ImageNet) + frozen BERT base.
- Projection heads + 256-d shared space, learnable temperature.
- Symmetric InfoNCE, AdamW.
- Single fixed prompt; stratified 80/10/10 split (Table 3.1).

4.2.2 Training Behaviour

- Train/val loss + temperature trajectory (figure).

4.2.3 Classification Performance

- Global metrics (Table 4.2).
- Per-class P/R/F1 (Table 4.3).

Table 4.1 Experiment matrix grouped by axis of variation..

Experiment	Image enc.	Text enc.	Training	Axis tested
Reference points (no LC25000 training)				
PLIP zero-shot	ViT-B/16 (PLIP, frozen)	CLIP-text (PLIP, frozen)	none	pathology pretraining
OpenAI CLIP zero-shot	ViT-B/16 (frozen)	CLIP-text (frozen)	none	architecture-controlled
Trained on LC25000				
Baseline	ResNet50 frozen	BERT frozen	CLIP InfoNCE	reference
Plain image classifier	ResNet50 frozen	—	cross-entropy	value of CLIP paradigm
Adaptation ablations				
Partial fine-tuning	ResNet50 last-30 unfrozen	BERT frozen	CLIP InfoNCE	feature adaptation
Stain normalization	ResNet50 frozen	BERT frozen	CLIP InfoNCE	stain shortcut / method head-to-head
Text-side ablation				
Biomedical text encoder	ResNet50 frozen	BioBERT / PubMedBERT frozen	CLIP InfoNCE	biomedical text pretraining
Generalization				
External colon dataset	ResNet50 frozen	BERT frozen	baseline checkpoint	OOD transfer
Image encoder family				
DINOv2 image encoder	DINOv2 frozen	BERT frozen	CLIP InfoNCE	self-supervised pretraining

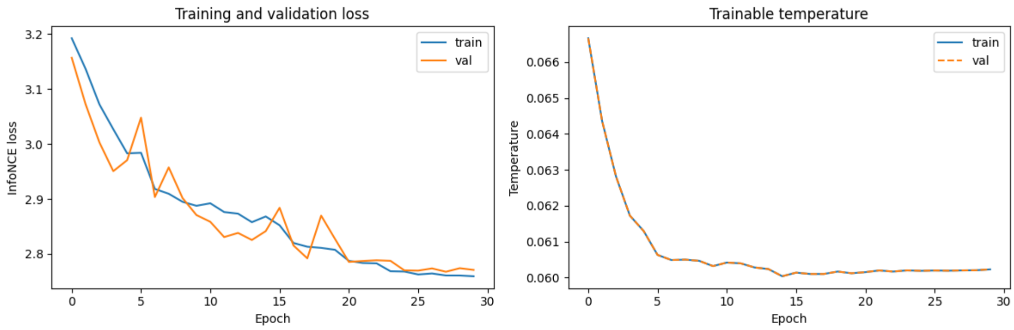


Figure 4.1 Baseline training and validation InfoNCE loss (left) and learnable temperature τ (right) against epoch. τ drifts from the CLIP initialisation of 0.07 toward its converged value..

Table 4.2 Baseline global classification metrics on the test set..

Metric	Value
Accuracy	0.6356
Balanced accuracy	0.6356
Macro F1	0.6273
Weighted F1	0.6273

4.2.4 Confusion Matrices

- Raw + row-normalized (figure pair).

4.2.5 Embedding Visualizations

- Shared image+prompt UMAP (two views: vanilla scatter and sample-image thumbnail overlay).
- Centroid-to-prompt similarity heatmap (figure).

Table 4.3 Baseline per-class classification metrics..

Class	Precision	Recall	F1	Support
benign lung tissue	0.639	0.702	0.669	500
lung adenocarcinoma	0.524	0.442	0.479	500
lung squamous cell carcinoma	0.705	0.840	0.766	500
benign colon tissue	0.630	0.722	0.673	500
colon adenocarcinoma	0.656	0.472	0.549	500
Macro average	0.631	0.636	0.627	2 500

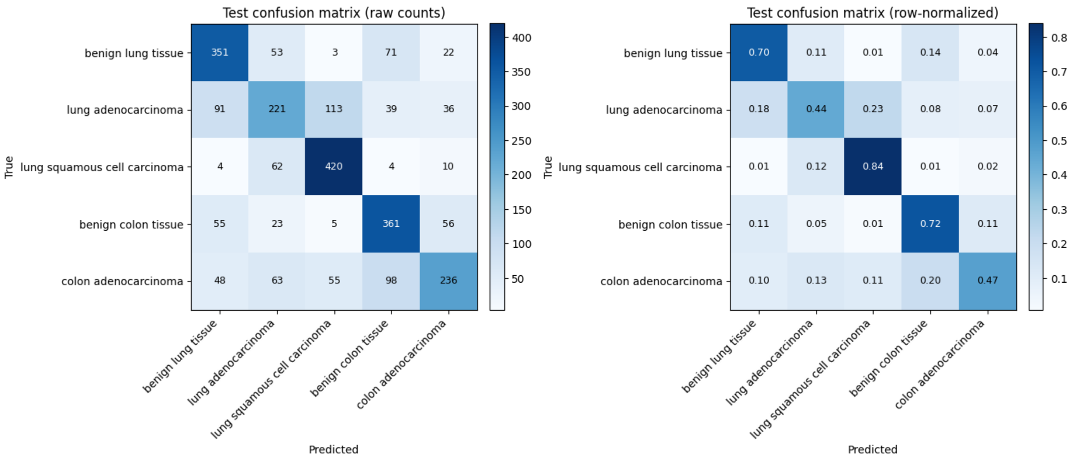
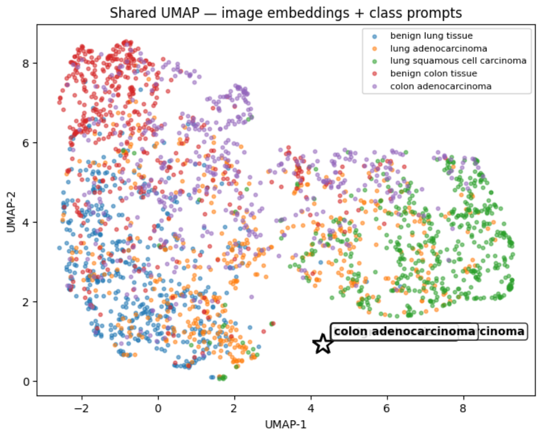
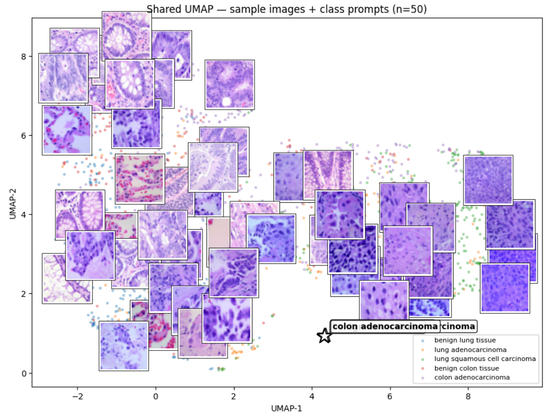


Figure 4.2 Baseline confusion matrices. Left: raw counts. Right: row-normalised by true class..



(a) Vanilla scatter.



(b) Sample-image thumbnails.

Figure 4.3 Shared UMAP of test image embeddings and the five class-prompt embeddings (cosine metric), in two views from the same fit. (a) Per-class image scatter with class-prompt embeddings marked. (b) Same coordinate system overlaid with ∼50 sample image thumbnails (no two within a minimum distance) and prompt markers on top — adapted from the reference notebook supplied by the tutor..

4.2.6 Class-Level Retrieval

- Positives = class-shared (not per-sample). Not CLIP-paper retrieval.
- R@K + P@K (table).

4.2.7 Baseline Interpretation

- Class-by-class confusion structure: lung_aca worst, lung_scc best.

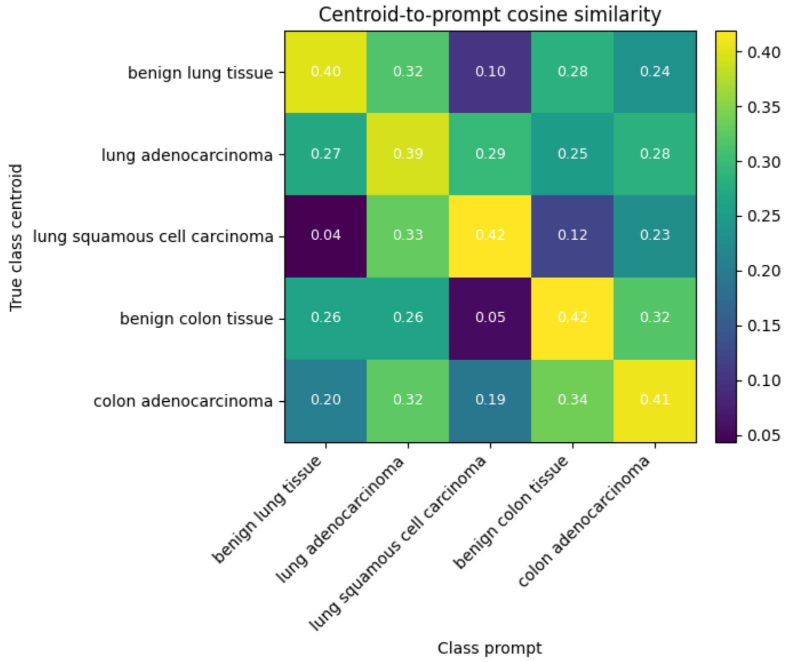


Figure 4.4 Baseline centroid-to-prompt cosine similarity heatmap. Rows: image-embedding centroid for each true class. Columns: the five class-prompt embeddings. Cell values: cosine similarity. A diagonal-heavy matrix indicates that each class's images align most strongly with their own prompt..

Table 4.4 Baseline class-level retrieval..

Metric	K=1	K=5	K=10	K=50
Image → Class prompt R@K	0.6356	1.0000	—	—
Class prompt → Image P@K	0.8000	0.7200	0.8000	0.8400

- Stain artifacts → how much they plausibly contribute.
- Embedding structure → what it says about ImageNet+BERT → pathology transfer.

4.3 PLIP zero-shot comparison

4.3.1 Setup

- PLIP (Huang 2023): CLIP on ∼200k pathology Twitter pairs.
- Loaded `vinid/plip`, evaluated zero-shot on same test split + prompts.
- Isolates pathology-pretraining contribution.

4.3.2 Headline numbers vs baseline

Table 4.5 Pathology pretraining contribution to frozen CLIP transfer..

Metric	Baseline (general)	PLIP (pathology)	Δ
Accuracy	0.6356	0.6980	+0.062
Balanced accuracy	0.6356	0.6980	+0.062
Macro F1	0.6273	0.7004	+0.073
Weighted F1	0.6273	0.7004	+0.073

4.3.3 Per-class

- Big gains on benign tissues (lung F1 0.67→0.87; colon F1 0.67→0.80).
- Cancer subtypes essentially tied (lung_aca F1 0.48→0.48; colon_aca F1 0.55→0.56).
- Lung SCC recall jumps (0.84→0.93), precision drops (0.70→0.68) → overcalls SCC.

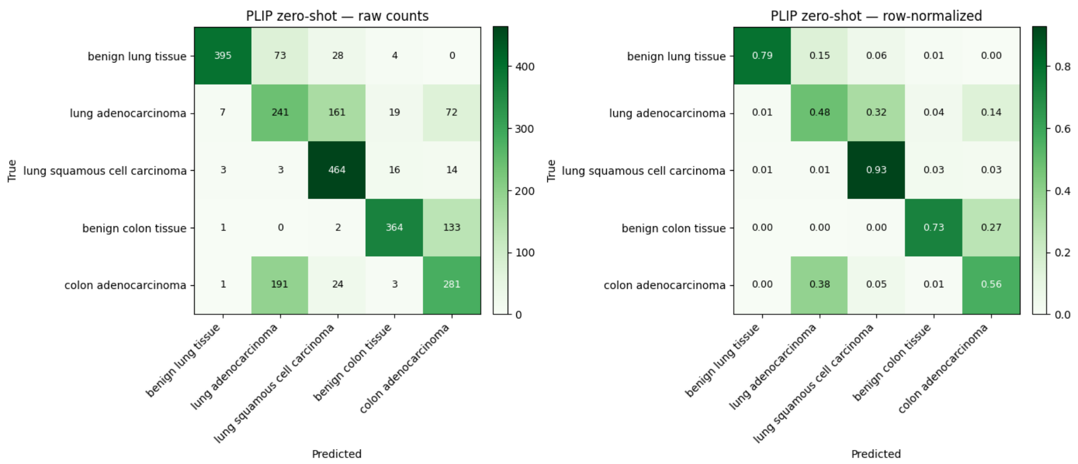
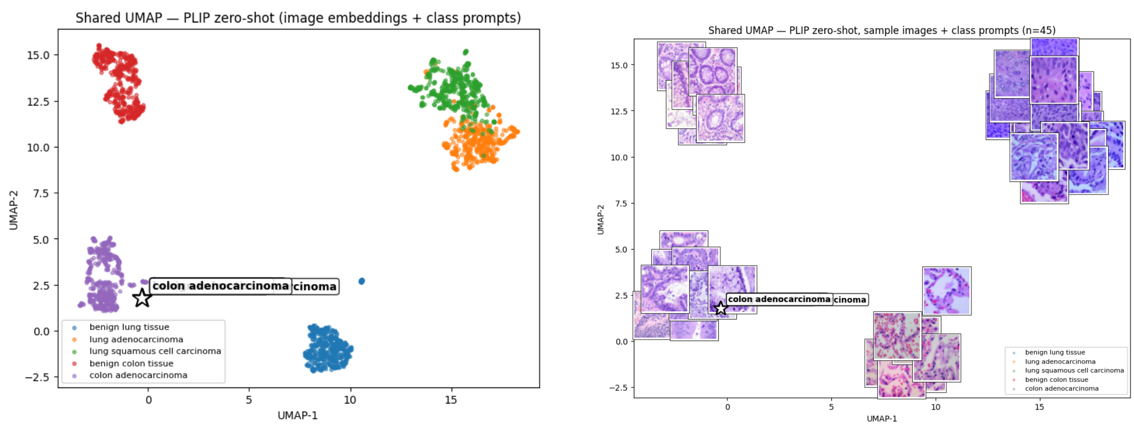


Figure 4.5 PLIP zero-shot confusion matrices, same five classes as the baseline. Left: raw counts. Right: row-normalised by true class..



(a) Vanilla scatter. (b) Sample-image thumbnails.

Figure 4.6 PLIP zero-shot shared UMAP of test image embeddings and the five class-prompt embeddings (cosine metric). Image clusters separate by class meaningfully better than the baseline (Figure 4.3), but the five prompt markers still collapse into a near-singular point — the same short-prompt + frozen text encoder degeneracy as the baseline. PLIP earns its +7.3 macro-F1 by producing better image embeddings, not by separating prompts..

4.3.4 Retrieval contrast

- I→T R@1 = 0.6980 (matches accuracy).
- T→I P@K *worse* than the baseline (P@1: 0.40 vs 0.80; P@50: 0.53 vs 0.84).
- Reading: the baseline was trained on LC25000 → embedding space tuned for in-distribution ranking. PLIP zero-shot → wins classification, loses fine ranking.

4.3.5 Interpretation

- Pathology pretraining ≈ +7.3 macro-F1 on frozen CLIP transfer.
- Gain in benign-vs-malignant, not in subtype discrimination.

- Image clusters in the shared UMAP separate visibly better than the baseline (Figure 4.6), but the prompt markers still collapse — domain pretraining helps the image side, not the (still-degenerate) text side.
- Feature adaptation confirmed as the bigger lever in §4.5; PLIP's frozen-domain gain is one-third of unfreezing's frozen-vs-adapted gain.

4.4 Plain image classifier on the same features

- Drops the CLIP architecture: no text encoder, no projection heads, no learnable temperature, no contrastive loss.
- Replaces with: ResNet50 $\rightarrow$ GAP $\rightarrow$ BatchNorm $\rightarrow$ Dense(5) trained with sparse cross-entropy.
- Run at both feature-adaptation budgets (frozen and unfreeze30) so the comparison vs the CLIP variants in §4.2 and §4.5 is a complete 2×2.
- Isolates what the CLIP architecture itself contributes once the image encoder, data and adaptation budget are held constant.

Table 4.6 CLIP vs plain CE classifier on the same ResNet50 backbone at each feature-adaptation budget. Frozen regime: tied within noise. Unfreeze30 regime: CLIP wins +2.6 macro-F1..

	Frozen		**Unfreeze last 30**	
Architecture	Acc.	Macro F1	Acc.	Macro F1
CLIP (InfoNCE)	0.6356	0.6273	**0.8984**	**0.8981**
Plain (CE softmax)	0.6440	0.6369	0.8728	0.8723
Δ (CLIP – plain CE)	−0.0084	−0.0096	+0.0256	+0.0258

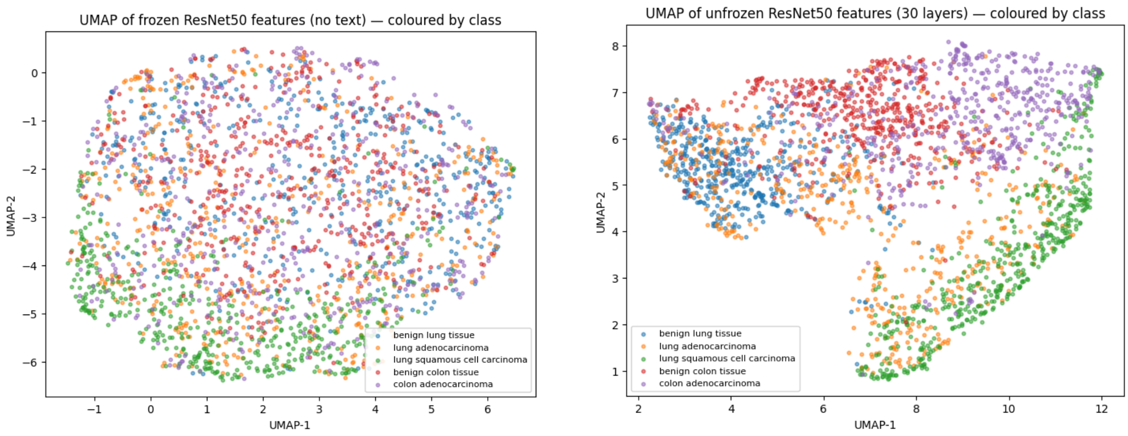
4.4.1 Interpretation

- Frozen regime: CLIP and plain CE tie within noise. Both saturate at the ImageNet-features ceiling of ~ 0.63; the architecture choice is invisible when the feature space is fixed.
- Unfreeze30 regime: CLIP outperforms plain CE by +2.6 macro-F1, broad-based across every class (benign lung +0.018; lung_aca +0.022; lung_scc +0.020; benign colon +0.038; colon_aca +0.031).
- Plausible mechanism: InfoNCE with five shared class-prompt targets acts as implicit regularisation against per-image memorisation. With ~ 14 M trainable parameters in the unfrozen tail, requiring every sample of a class to align with the same prompt limits how much sample-specific information the model can store. The benefit only materialises once the model has the capacity to overfit, which is why the frozen regime sees no difference.
- The CLIP paradigm's value on a closed-set 5-class task is not raw accuracy. The +2.6 macro-F1 bonus under feature adaptation is real but secondary — the paradigm's load-bearing contribution is capabilities a plain classifier cannot provide: open-vocabulary classification at inference, image–text retrieval, prompt-tuning as a controlled lever.

4.5 Feature-level adaptation: partial unfreezing

- Same arch/loss/prompts/data/split as the baseline; only the trainable mask on ResNet50 changes.
- Last 30 of 178 ResNet50 layers unfrozen (≈ 14.4 M extra trainable params on top of the projection heads).
- BatchNorm in the unfrozen tail updates its running stats during training (training mode) and uses them at eval.
- LR dropped $10\times$ to 10^{-4} to protect the pretrained features.

- +27.1 macro-F1 over the baseline ($\approx 3.7\times$ the PLIP zero-shot gain).



(a) Frozen ResNet50 features.

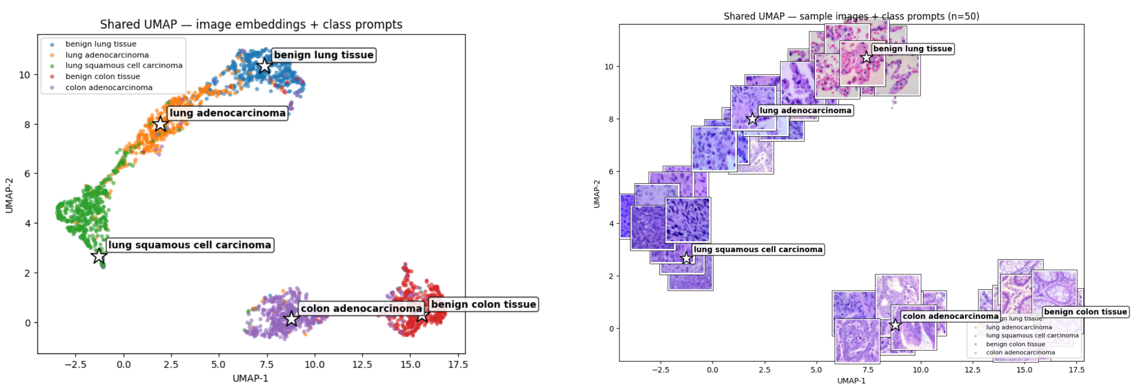
(b) Last 30 layers unfrozen.

Figure 4.7 Plain-classifier UMAP of test image features, coloured by true class. No text encoder in this ablation, so no prompt markers — this is an image-side view only. (a) Frozen features show weak, tangled class structure consistent with the ~ 0.64 classification ceiling. (b) With the last 30 layers unfrozen, classes resolve into five well-separated clusters, consistent with the ~ 0.87 accuracy..

Table 4.7 Partial fine-tuning vs frozen baseline: global classification metrics on the held-out test set..

Metric	Baseline	Unfreeze last 30
Accuracy	0.6356	**0.8984**
Balanced accuracy	0.6356	**0.8984**
Macro F1	0.6273	**0.8981**
Weighted F1	0.6273	**0.8981**

- Broad-based across classes: every class $F_1 \geq 0.85$.
- Lung_aca: F_1 0.48 $\rightarrow$ 0.85 (worst baseline class; still hardest but no longer a failure mode).
- $+0.25$ macro-F1 over the frozen-features ceiling measured by the plain CE classifier (~ 0.64, see §4.4). Headroom to fully-fine-tuned literature results on LC25000 (~ 0.95–0.99, end-to-end training, often with stronger backbones) is harder to pin down precisely without running that regime ourselves.



(a) Vanilla scatter.

(b) Sample-image thumbnails.

Figure 4.8 Unfreeze30 shared UMAP of test image embeddings and the five class-prompt embeddings (cosine metric), in two views from the same fit. Compare with the baseline UMAP in Figure 4.3 — the unfrozen-tail embedding space shows tighter, better-separated class clusters and prompt markers that sit clearly inside their own cluster..

4.6 Stain normalization ablation

- Same arch/loss/prompts/split as the baseline; only the image inputs change.
- Head-to-head: Macenko, Reinhard, Vahadane on the same fixed reference patch.
- Per-method delta isolates the stain-normalization method.
- Result: *TBD*.

4.7 Biomedical text encoder ablation

- Same arch/loss/prompts/split as the baseline; only the text encoder changes.
- Swap BERT base $\rightarrow$ BioBERT, then $\rightarrow$ PubMedBERT (frozen in both cases).
- Tests whether biomedical text pretraining helps under short class-name prompts.
- Result: *TBD*.

4.8 Out-of-distribution generalization

- Zero-shot eval of LC25000-trained variants on an external colon dataset.
- Includes PLIP zero-shot as a domain-pretrained reference.
- Result: *TBD*.

4.9 Image encoder family

- Same text encoder/prompt/split/loss/metrics; image encoder swapped.
- DINOv2 and/or OpenAI CLIP ViT-B/16; supervised vs self-supervised pretraining.
- Deferred: not run in this project.

4.10 Discussion

- Synthesis across matrix rows as they land.
- Mapping back to RQ1–RQ7 with numbers.
- Frozen general-purpose ResNet50 features establish the reference floor for transfer.

Table 4.8 Summary of all experiments..

Experiment	**Accuracy**	**Macro F1**	**Δ vs baseline**
Baseline	0.6356	0.6273	—
PLIP zero-shot	0.6980	0.7004	+0.073
Plain image classifier (frozen)	0.6440	0.6369	+0.010
Plain image classifier (unfreeze30)	0.8728	0.8723	+0.245
Partial fine-tuning (CLIP, unfreeze30)	0.8984	0.8981	+0.271
Stain normalization (best of M/R/V)	—	—	—
Biomedical text encoder (best)	—	—	—
External colon dataset	—	—	—

5 Conclusions and Future Work

5.1 Conclusions

- Baseline result (RQ1): 63.6% / 62.7% macro-F1, worst class lung adenocarcinoma.
- Pathology pretraining (RQ2): PLIP $+7.3$ macro-F1, mainly benign vs malignant.
- Feature adaptation (RQ3): partial unfreezing (last 30 of 178 layers) $+27.1$ macro-F1, broad-based across classes.
- CLIP-paradigm value (RQ4): CLIP and plain CE tie at the frozen-features ceiling (~ 0.63); CLIP wins $+2.6$ macro-F1 over CE under feature adaptation (unfreeze30), broad-based across all classes. The paradigm's primary value remains in capabilities (open-vocabulary, retrieval), not raw accuracy.
- Stain normalization (RQ5): *TBD*.
- Biomedical text encoder (RQ6): *TBD*.
- OOD transfer (RQ7): *TBD*.
- Cross-cutting observations: saturation of frozen transfer, role of dataset artifacts.

5.2 Limitations

- Single-institution dataset, no patient-level grouping, no clinical validation.
- Small batch + few classes $\rightarrow$ InfoNCE false negatives, documented not mitigated.
- Frozen-backbone baseline tests transfer, not full fine-tuning.
- No human-expert evaluation.
- No head-to-head against CONCH / BiomedCLIP / MUSK (compute scope).

5.3 Future Work

Near-term:

- Additional external datasets beyond colon-Kaggle.
- SupCon or class-balanced sampling for false-negative regime.
- Patient-level grouping if metadata recoverable.

Not run in this project (low expected Δ under frozen text encoders + short prompts):

- Inference-time prompt ensembling.
- Dynamic prompt training.

Medium-term:

- Larger effective batch (grad accumulation or bigger GPU).
- Prompt-tuning methods (CoOp, CoCoOp).

Longer-term:

- Architecture-controlled comparison against CONCH / BiomedCLIP / MUSK / PathChat.
- Multi-institution training and evaluation.
- Returning to segmentation with pathology FM backbones.
- CLIP-with-short-labels applied to other low-label scientific domains.

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